

Requirement Analysis – Technology Stack (Detailed)

Project: Food Survey Analysis using Tableau and Flask

1. Introduction

The Technology Stack document explains the software tools, programming languages, frameworks, libraries, and platforms used to design, develop, test, and deploy the Food Survey Analysis project. The selected technologies provide efficient data processing, interactive visualization, and easy project maintenance.

2. Purpose

The objective of the technology stack is to ensure that every component of the application works together efficiently while providing scalability, maintainability, and high-quality visual analytics.

3. Frontend Technologies

HTML structures the web pages, CSS provides styling, and JavaScript enables interactive elements. These technologies are used in the Flask templates to display the Tableau dashboard and project pages.

4. Backend Technologies

Python is the primary programming language. Flask is used as the lightweight web framework to host the application and embed the Tableau dashboard.

5. Data Analytics Technologies

Tableau Public/Desktop is used to create worksheets, dashboards, KPI cards, stories, histograms, pie charts, scatter plots, treemaps, box plots, and other visualizations. Pandas supports data preparation and analysis before visualization.

6. Version Control

Git manages source code history, while GitHub stores the project repository, documentation, screenshots, and project reports, enabling collaboration and version tracking.

7. Development Tools

Visual Studio Code is used for coding. Python virtual environments manage dependencies. Modern web browsers such as Google Chrome and Microsoft Edge are used for testing the application.

8. Deployment

The completed Tableau dashboard can be published on Tableau Public, while the Flask application can be executed locally or deployed to a compatible hosting platform.

Technology	Purpose
Python	Backend programming
Flask	Web application framework
Tableau	Interactive dashboards and analytics
Pandas	Data preparation and processing
HTML	Web page structure
CSS	User interface styling
JavaScript	Interactive behaviour
Git	Version control
GitHub	Repository hosting
VS Code	Development environment

9. Benefits

The selected technology stack provides reliable performance, interactive data visualization, easy maintenance, cross-platform compatibility, scalability, and efficient project development.

10. Conclusion

The chosen technology stack satisfies the requirements of the Food Survey Analysis project by combining Python, Flask, Tableau, and supporting web technologies into a complete analytics solution capable of delivering meaningful insights.